SHETH L.U.J AND SIR M.V. COLLEGE OF ARTS SCIENCE AND COMMERCE

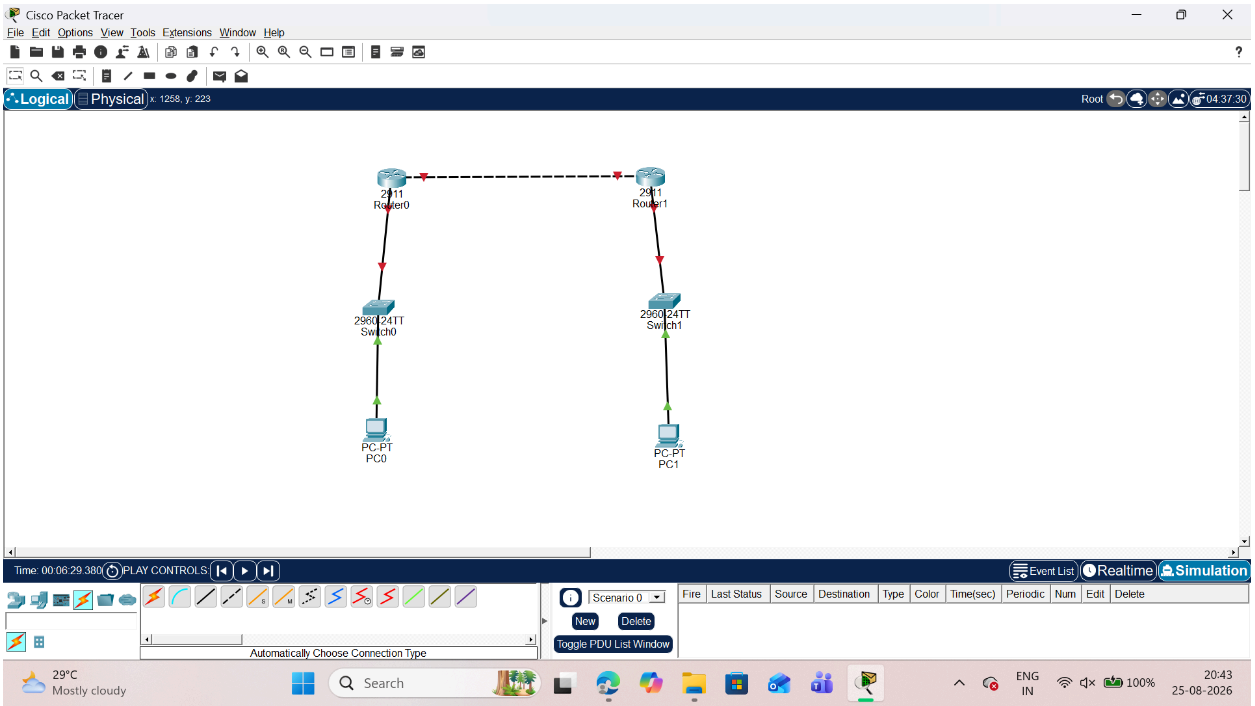
SUBJECT: CYBER & INFORMATION SECURITY

PRACTICAL - 6

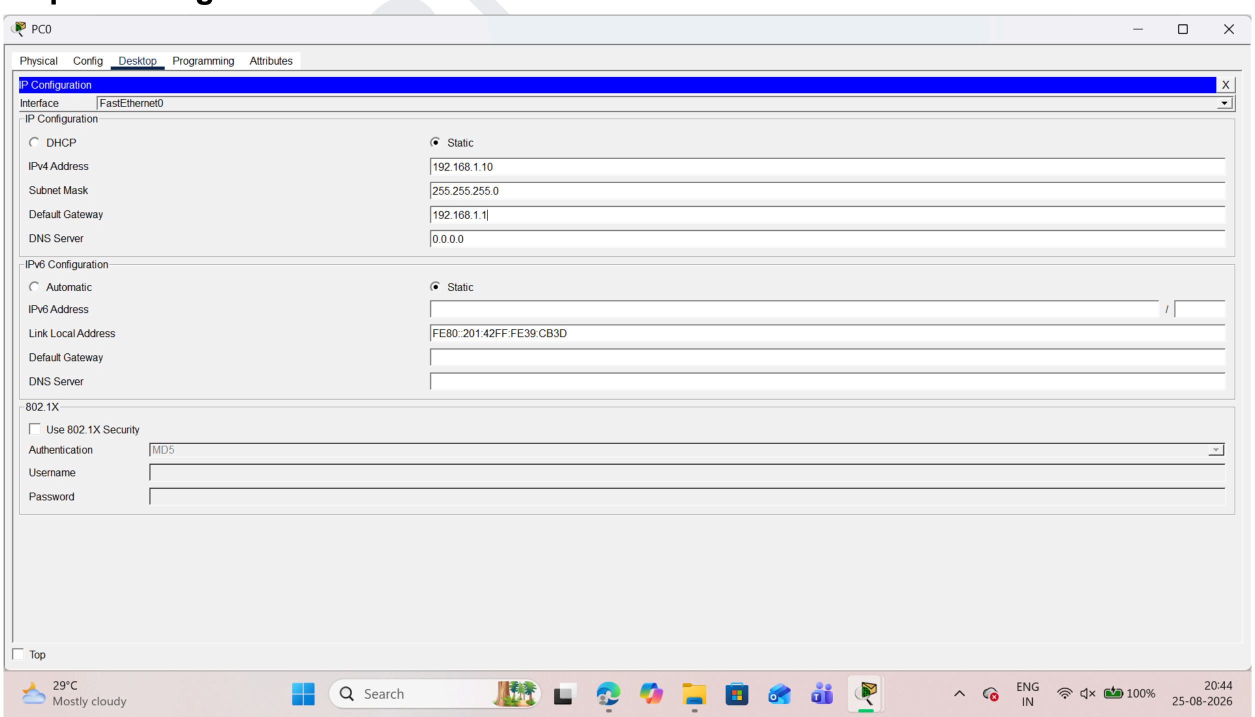
AIM:IP Security (IPsec) Configuration

Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Step 1. Network topology



Step 2. Configure PC IP addresses a.PC0



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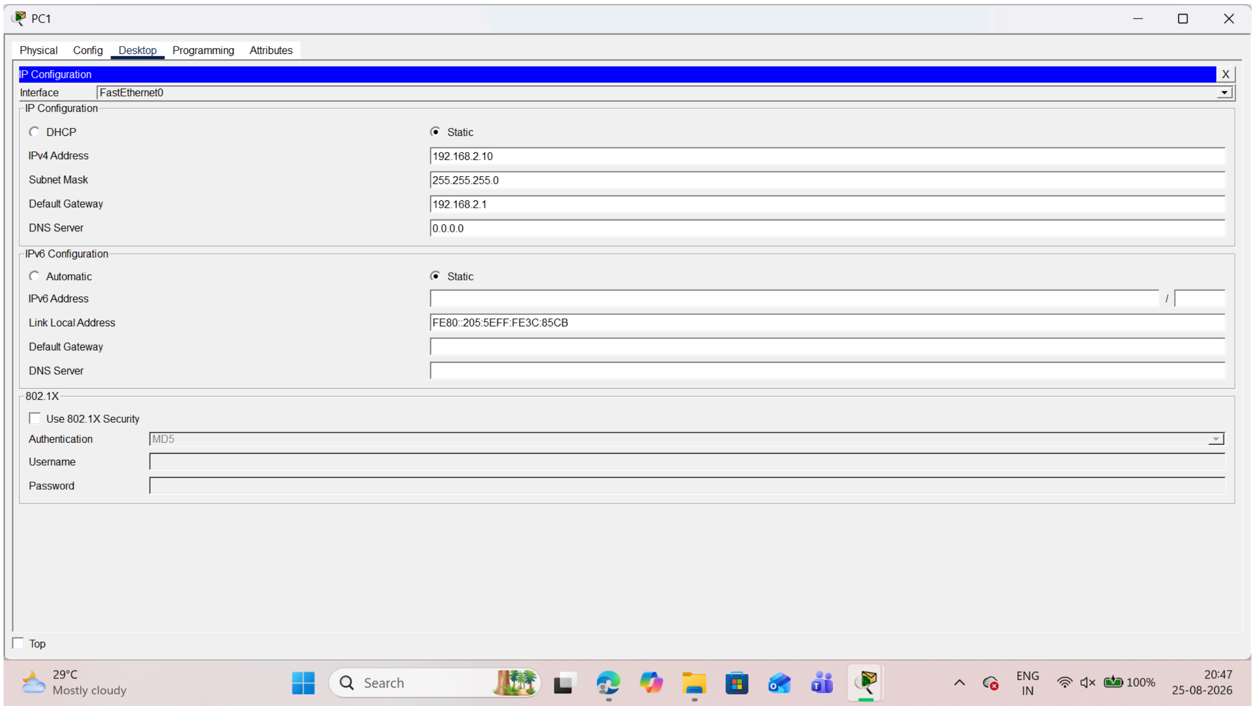
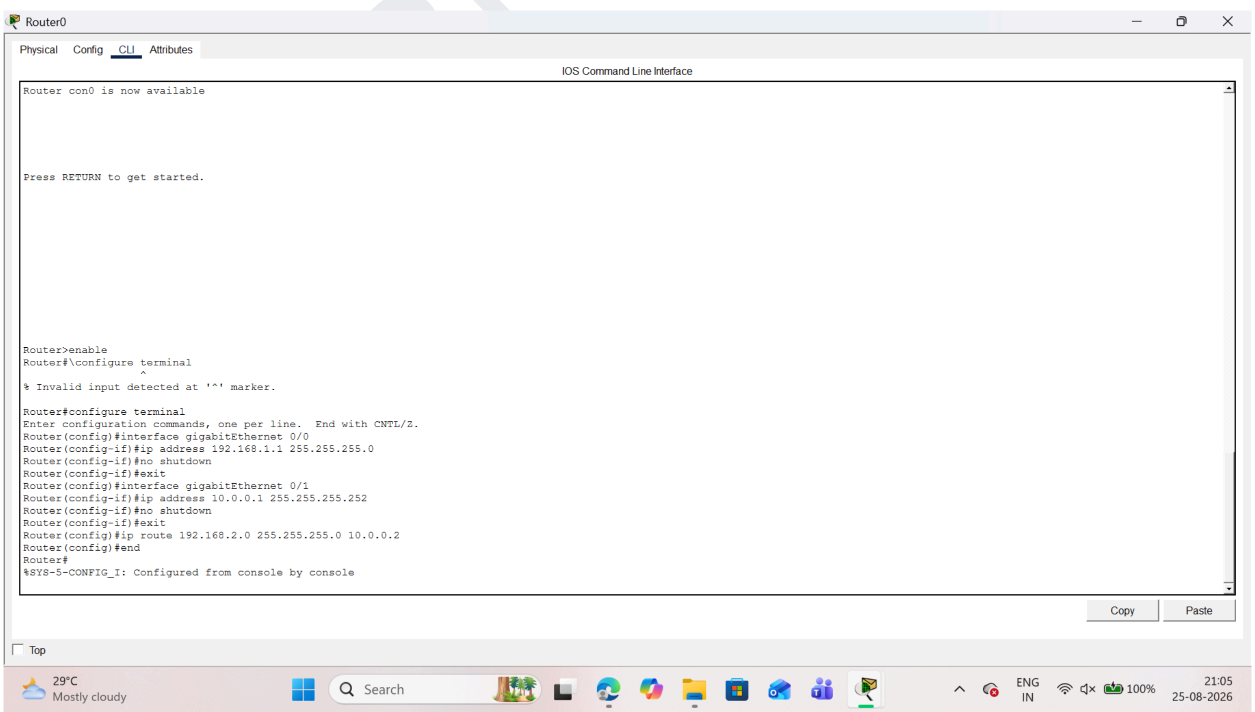
ROLL NO : T085

b.PC1



Step 3. Configuring the Routers

a.Router 0



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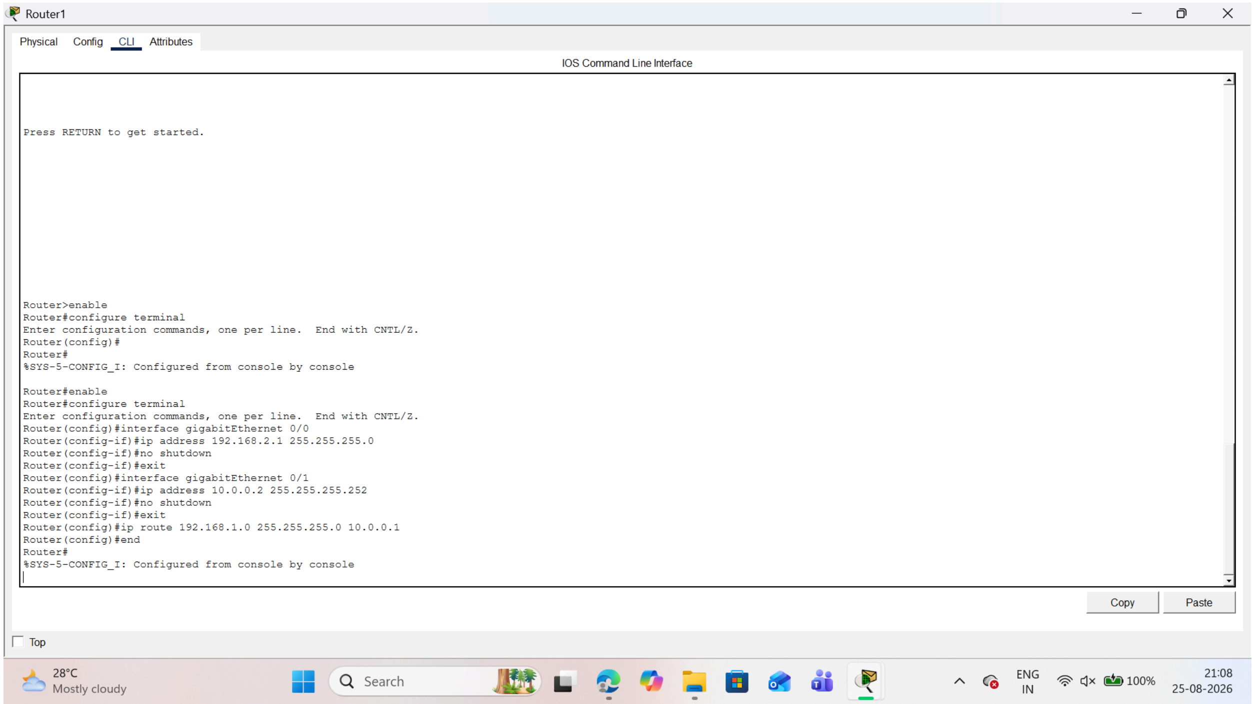
b.Router 1

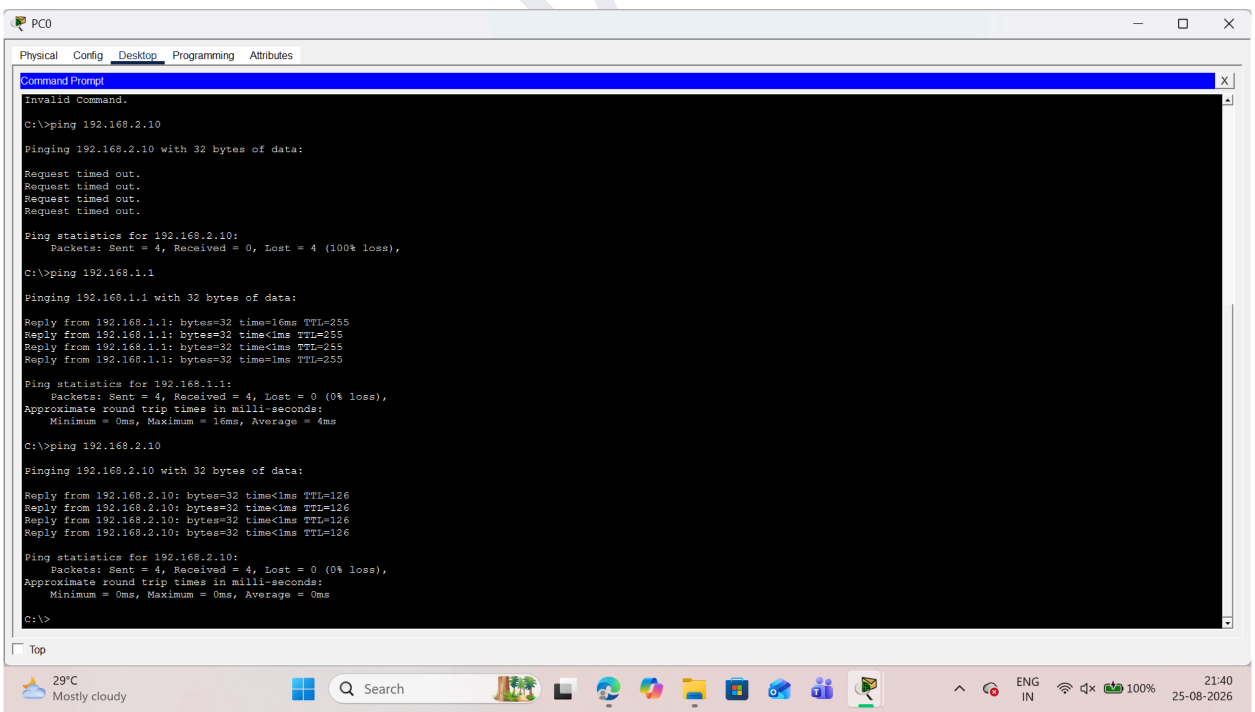


```
Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 4. First test connectivity using ping command



```
Invalid Command.

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=16ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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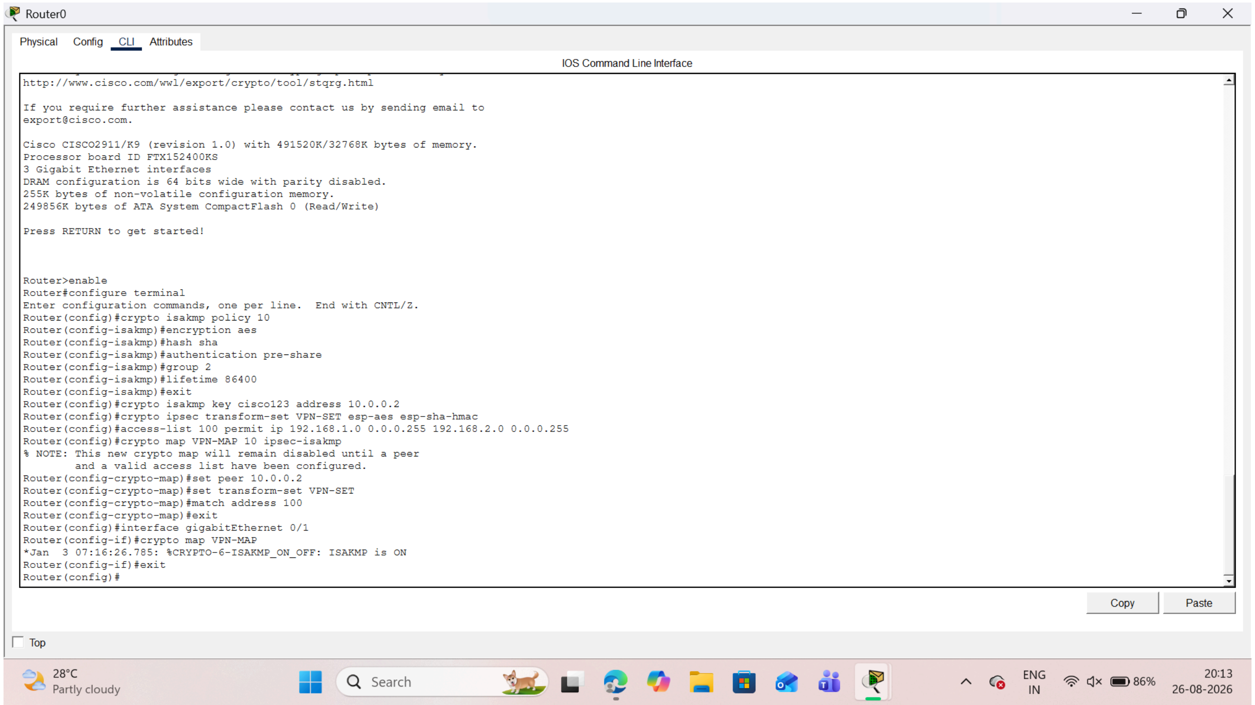
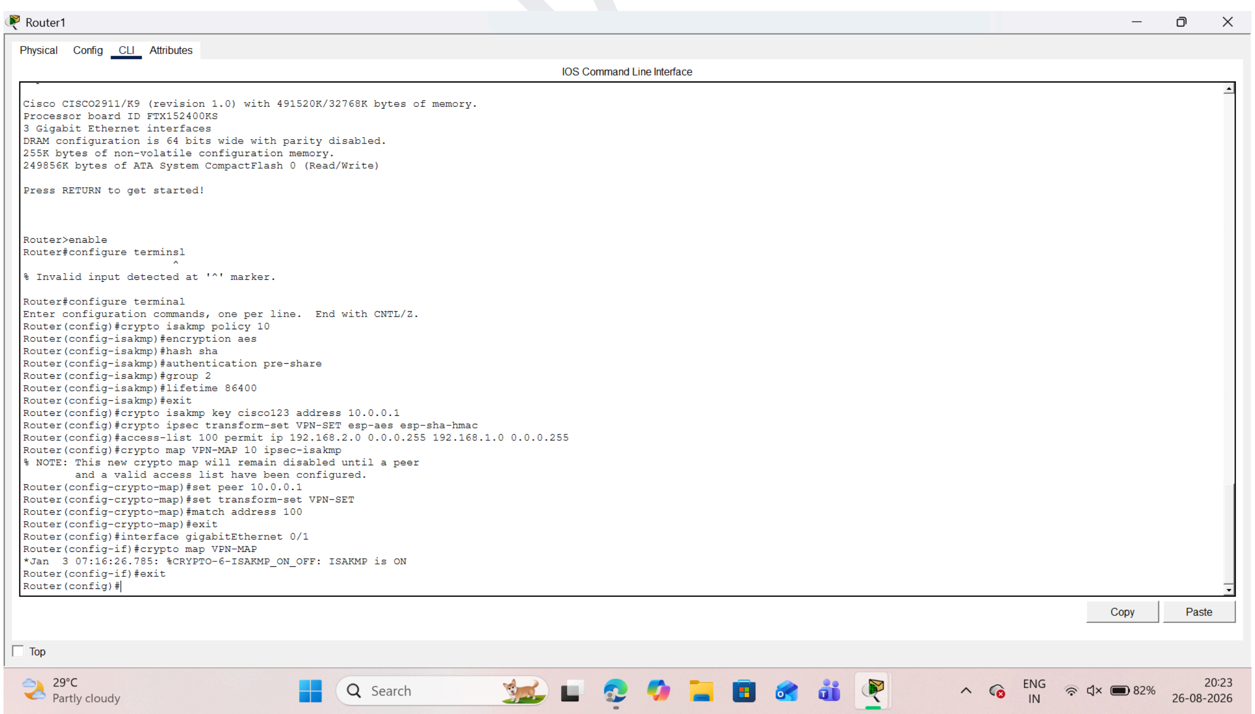
Step 5 . Configure IPsec on Router 0



Step 6 Configure IPsec on Router 1



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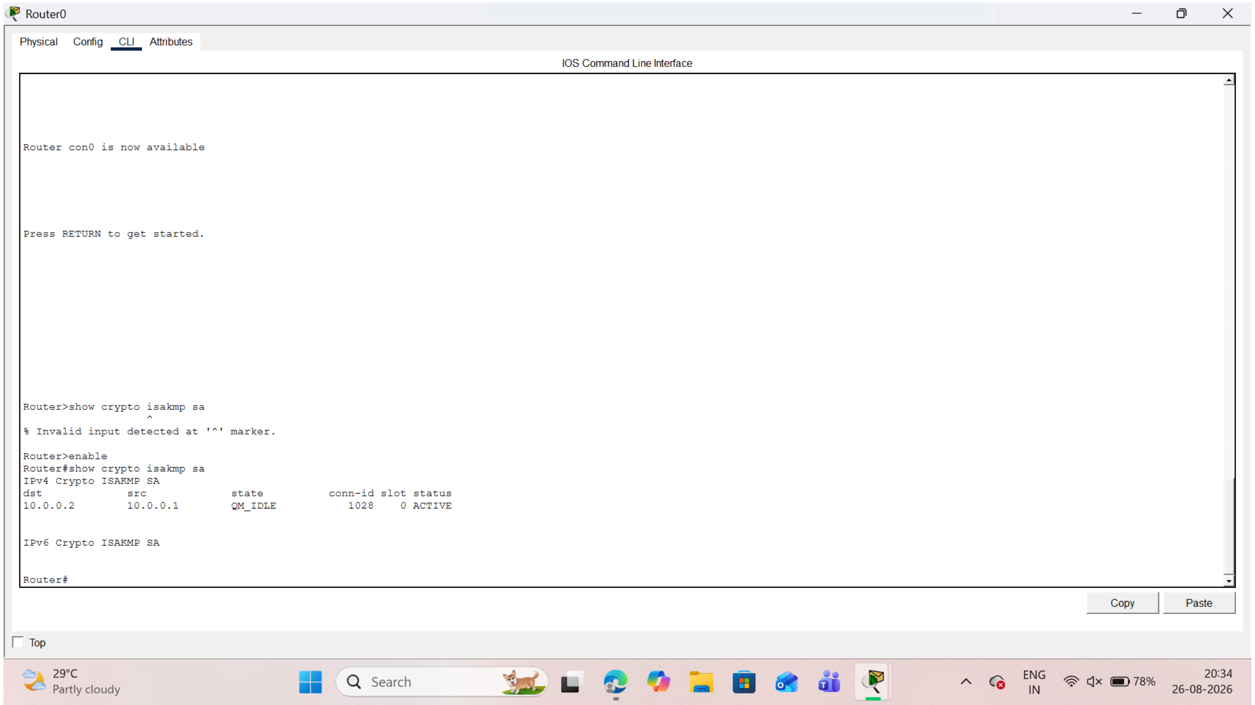
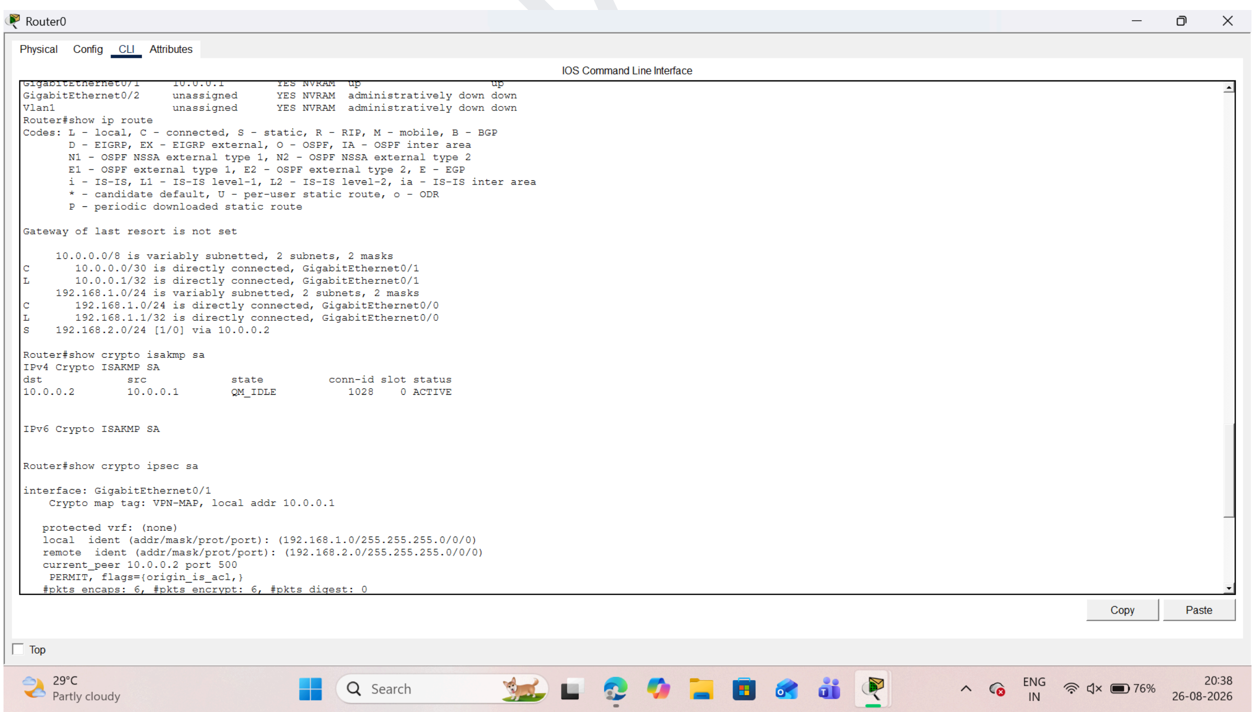
Step 7 Test the IPsec tunnel and verifying





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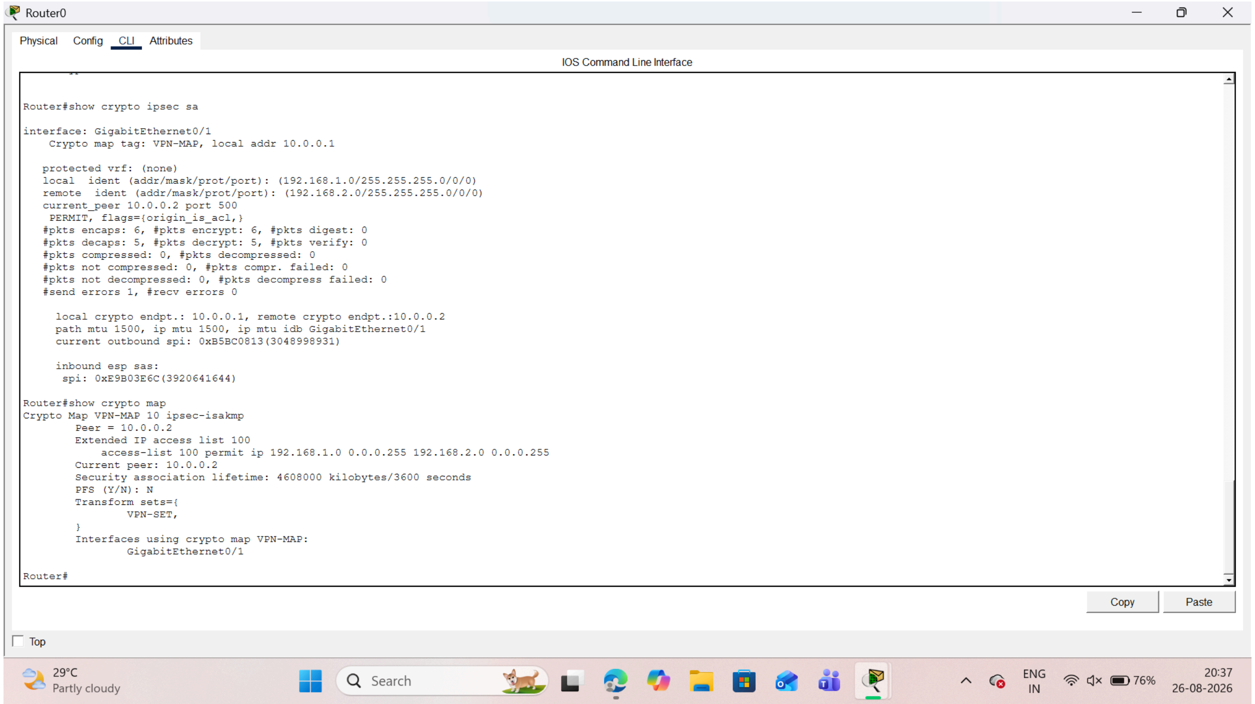

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